

Assignment 3

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1 Question 1:

In this section, I iterated the function $z_{n+1} = z_n^2 + c$ and the set of all points c in the complex plane for which the sequence remains bounded is in the middle of the plot. The central yellow shade and circular bulbs represent complex points c that are bounded and belong to the set. This was created by creating grid of points representing the complex plane, and for each point c , the sequence was iterated to determine if it remained bounded. The iteration continued until either the magnitude $|z_n|^2$ exceeded 4 or a maximum iteration count of 500 was reached. Then, it was plotted using matplotlib.

2 Question 2: The Lorenz Equations

The following systems were solved using scipy's `solve_ivp`.
 $\dot{X} = -\sigma(X - Y), \quad \dot{Y} = rX - Y - XZ, \quad \dot{Z} = -bZ + XY$ (1) Using the parameters $\sigma = 10, r = 28$, and $b = 8/3$.

The system was integrated for dimensionless time units $t = 60$ using an ODE solver with the original Lorenz parameter values: $\sigma = 10, r = 28$, and $b = 8/3$. The initial conditions used were $W_0 = [0., 1., 0.]$. Later, to analyze sensitivity, a second integration was performed with a slightly perturbed initial condition, $W'_0 = [0., 1.00000001, 0.]$. Figure ?? shows the trajectory in the $Y-Z$ phase plane between $t = 14$ and $t = 19$.

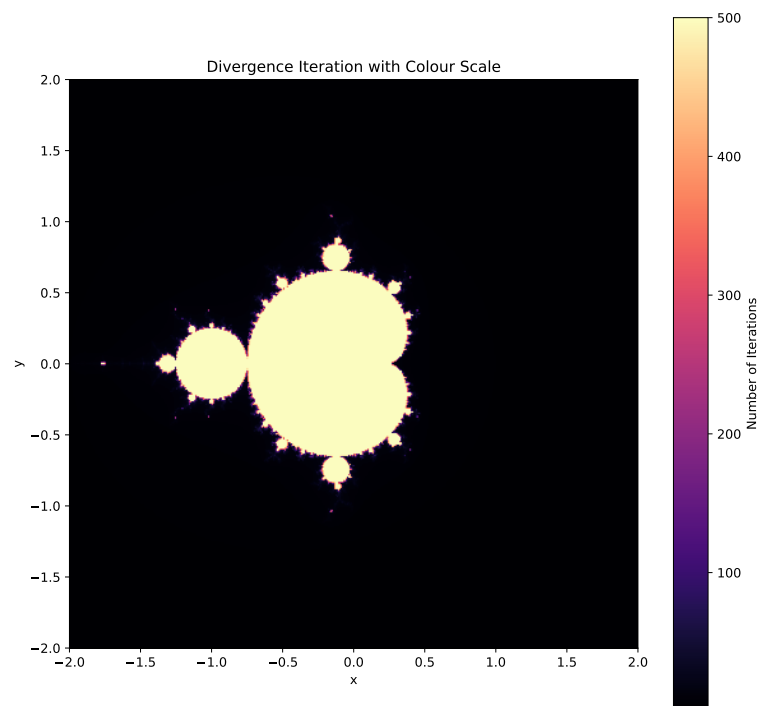


Figure 1: The set visualized with a color scale representing the rate of divergence.

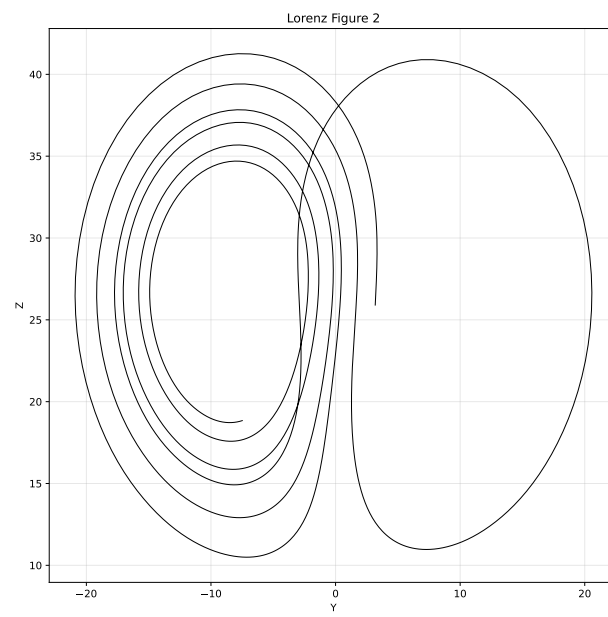


Figure 2: Evolution of variable Y against $N = t/0.01$.